

Sample data and Test Cases

Script Generation Test Cases

Test Case 1

Input

```
{  
  "software_name": "Docker",  
  "software_version": "latest",  
  "os_name": "Ubuntu",  
  "os_version": "22.04",  
  "package_manager": "apt"  
}
```

Expected Output

```
sudo apt update  
sudo apt install docker.io -y  
sudo systemctl enable docker  
sudo systemctl start docker
```

Test Case 2

Input

```
{  
  "software_name": "Python",  
  "software_version": "3.12",  
  "os_name": "Windows",  
  "os_version": "11",  
  "package_manager": "winget"  
}
```

Expected Output

```
winget install Python.Python.3.12
```

Test Case 3

Input

```
{  
  "software_name": "Node.js",  
  "software_version": "20",  
  "os_name": "Ubuntu",  
  "os_version": "22.04",  
  "package_manager": "apt"  
}
```

Expected Output

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -  
sudo apt install -y nodejs
```

3. Compatibility Validation Test Cases

Valid Configuration

Input

```
{  
  "software": "Docker",  
  "os": "Ubuntu 22.04"  
}
```

Expected Output

```
{  
  "compatible": true  
}
```

Invalid Configuration

Input

```
{  
  "software": "Docker",  
  "os": "Windows XP"  
}
```

Expected Output

```
{  
  "compatible": false,  
  "message": "Unsupported operating system"  
}
```

4. Troubleshooting Agent Test Cases

Input

```
{  
  "error_message": "docker daemon not running"  
}
```

Expected Output

```
{  
  "root_cause": "Docker service stopped",  
  "solution": "sudo systemctl restart docker",  
  "confidence": 95  
}
```

Input

```
{  
  "error_message": "python is not recognized as an internal or external command"  
}
```

Expected Output

```
{  
  "root_cause": "Python not added to PATH",  
  "solution": "Reinstall Python with Add to PATH enabled"
```

AI Assistant Test Cases

Question

How do I install Docker on Ubuntu?

Expected Response

Provides Docker installation commands and verification steps.

Question

How do I verify Python installation?

Expected Response

python --version

6. Report Generation Test Case

Input

```
{  
  "software_name": "Docker",  
  "os_name": "Ubuntu",  
  "os_version": "22.04"  
}
```

Expected Result

Generated PDF containing:

Installation Summary

Installation Script

Validation Results

Troubleshooting Guide

Timestamp

Demo Scenario for Interview

Login

admin@installgen.ai

Password@123

Request

Software: Docker

Version: Latest

OS: Ubuntu

OS Version: 22.04

Package Manager: apt

System Flow

Requirement Agent

↓

Environment Agent

↓

Knowledge Base Agent

↓

Script Generator Agent

↓

Validation Agent

↓

Troubleshooting Agent

↓

Documentation Agent

Output

sudo apt update

sudo apt install docker.io -y

sudo systemctl enable docker

sudo systemctl start docker

Plus:

Compatibility Status

Troubleshooting Guide

Downloadable Script

PDF Report

This gives you a complete set of realistic test cases and demo data for documentation, testing, and interview presentation.